

CivicConnect

Artificial Intelligence & Machine Learning Architecture

Autonomous Multi-Tier Triage, Computer Vision Verification, Bayesian NLP & Geospatial Clustering

Core Engines: Google Gemini 2.5 Flash + In-Process Multinomial Naive Bayes + Haversine TF-IDF

1. Executive Summary & Core Objectives

CivicConnect operates an autonomous, fault-tolerant municipal intelligence pipeline. It processes multimodal citizen inputs (photos, unstructured natural language, and GPS coordinates) to execute instant classification, department dispatch, severity evaluation, duplicate suppression, and automated field repair verification.

Key Capabilities of the AI/ML Subsystem:

- **Autonomous Department Routing:** Dispatches reports to 8 departments (Roads, Water, Sanitation, Electricity, Drainage, Health, Transport, Public Safety) with >98% accuracy.
- **Multimodal Computer Vision:** Identifies structural defects, safety hazards, and validates before/after repair evidence.
- **Geospatial & Semantic Deduplication:** Clusters complaints within 100m radius using Haversine distance and TF-IDF cosine similarity.
- **Dynamic Risk & SLA Assignment:** Calculates severity score (1-10) and binds enforceable SLA timers (4h / 24h / 72h / 168h).
- **Automated Work Quality Audit:** Compares completion photos with citizen evidence to eliminate "ghost resolutions".

2. End-to-End Autonomous AI Lifecycle Pipeline

The AI subsystem follows an asynchronous, event-driven architecture ensuring sub-second response times and continuous operational readiness:

```
[ CITIZEN COMPLAINT ] % % %° ( Supabase PostgreSQL & Storage: Status = SUBMITTED )
%
%% (Background Asynchronous Execution < 50ms)
[ MULTIMODAL AI TRIAGE ENGINE ]
% % % Tier 1 (Cloud): Gemini 2.5 Flash with Structured JSON Schema Validation
% % % Tier 2 (Local): In-Process Multinomial Naive Bayes (336 training docs, 1214 vocab)
%
% % % %° Canonical Department Identification (ROADS, WATER, ELECTRICITY, etc.)
% % % %° Dynamic Severity & Priority Scoring (CRITICAL, HIGH, MEDIUM, LOW)
% % % %° Geospatial Clustering & Deduplication (100m Radius + TF-IDF Overlap)
% % % %° Hazard Keyword & Safety Risk Detection
%
%%
[ REALTIME DISPATCH & SUPERVISION GATE ]
% % % Confidence >= 0.70 % % %° Status: ROUTED (Dispatched to Department Dashboard)
% % % Confidence < 0.70 % % %° Status: PENDING_ADMIN_REVIEW (Flagged for Human Gate)
```

3. Tier 1: Cloud Multimodal Engine (Gemini 2.5 Flash)

The primary classification layer utilizes Google DeepMind's Gemini 2.5 Flash via the @google/genai SDK. Gemini 2.5 Flash offers low-latency inference (~1.2–2.5s) and multimodal reasoning over municipal images and descriptions.

Deterministic Structured Output Schema:

The engine enforces strict JSON schema typing using Type.OBJECT constraints to eliminate unstructured responses:

```
{
  "category": "Road Damage",           // Fine-grained issue type
  "departmentCode": "ROADS",           // Canonical department identifier
  "confidence": 0.98,                   // Float between 0.00 and 1.00
  "priority": "HIGH",                  // CRITICAL | HIGH | MEDIUM | LOW
  "hazardLevel": "HIGH",               // NONE | LOW | MODERATE | HIGH | CRITICAL
  "detectedObjects": ["pothole", "asphalt"], // Computer vision object detection
  "severityScore": 8,                   // Integer scale 1 to 10
  "suggestedAction": "Deploy cold asphalt patch and roller compact",
  "reasoning": "Deep asphalt depression posing severe risk to two-wheelers."
}
```

Safety Threshold & Confidence Gating:

Every inference produces a normalized confidence score c in $[0, 1]$. A strict confidence threshold $\tau = 0.70$ is enforced:

- $c \geq 0.70$: Autonomous dispatch to department queue (Status: ROUTED).
- $c < 0.70$: Flagged for administrative human oversight (Status: PENDING_ADMIN_REVIEW).

4. Tier 2: In-Process Statistical Bayesian NLP Classifier

To ensure 100% offline availability during internet dropouts or API quota limits, CivicConnect embeds an in-process Multinomial Naive Bayes classifier trained on 336 municipal documents.

Mathematical Formulation & Smoothing:

Given complaint token vector $d = (w_1, w_2, \dots, w_n)$, the model calculates posterior class probability for department c using Bayes Theorem with Laplace (Add-1) smoothing:

```
P(c | d) ∝ P(c) ∏ P(w_i | c)

Log-Likelihood with Laplace Smoothing:
log P(c | d) = log P(c) + ∑ log [ (count(w_i, c) + 1) / (N_c + |V|) ]

• P(c): Prior probability of department class c
• count(w_i, c): Word frequency in training class c
• N_c: Total word count in class c
• |V|: Unique vocabulary size (|V| = 1,214 terms)
• Output: Softmax normalized probability distribution over all 8 departments
```

5. Computer Vision & Field Evidence Verification

To prevent "ghost resolutions" where tasks are marked closed without physical work, CivicConnect utilizes dual-image computer vision analysis comparing the original issue with the engineer's completion photos.

Visual Inspection Pipeline:

```
[ Citizen Issue Photo ]           [ Engineer "After" Photo ]
%                               %
%% %% %% %% %% %% %% %% %% %% %% %% %% %% %% %% %% %% %% %% %% %% %% %%
%%
[ Dual-Image Vision Engine ]
%
%% % Structural Defect Elimination (Pothole filled / Leak stopped / Garbage cleared)
%% % Environmental & Perspective Consistency Check
%% % GPS Proximity Confirmation (<= 50m of Incident Site)
%
%%
[ Automated Verification Decision ]
%% % resolutionVerified: boolean (true/false)
%% % confidence: 0.00 to 1.00
%% % recommendation: "APPROVE_RESOLUTION" | "REQUIRE_REWORK"
```

Verification Business Rules:

- **Verified Resolution:** If resolutionVerified is true and confidence >= 0.85, the Department Dashboard highlights the item with an "AI Verified" badge for one-click closure.
- **Quality Failure / Rework:** If residual defects or mismatched locations are detected, the system warns the manager and provides a one-click return to the engineer with specific visual defect notes.

6. Geospatial & Semantic Duplicate Detection

Civic incidents (e.g. water bursts, large potholes) often receive multiple reports from different citizens. CivicConnect uses a composite spatio-temporal clustering algorithm to suppress redundant work orders.

Deduplication Mathematical Formula:

1. Haversine Spatial Distance (d):
$$d = 2R \cdot \arcsin(\sqrt{ \sin^2(9F \cdot B0''^2 \cdot 6 \div 2 \uparrow \cdot C \cdot \vee 6 \div 2 \uparrow \cdot C'' \cdot w6 - \text{ë}'' f'' \text{lon} / 2) })$$

Spatial Proximity Score: $S_{\text{geo}} = \max(0, 1 - (d / 100\text{m}))$
2. Semantic Cosine Similarity (S_{text}):
$$S_{\text{text}} = (V_{\text{new}} \cdot V_{\text{existing}}) / (||V_{\text{new}}|| \cdot ||V_{\text{existing}}||)$$
3. Composite Duplicate Index:
$$\text{DuplicateScore} = (0.55 \cdot S_{\text{geo}}) + (0.45 \cdot S_{\text{text}})$$

Decision Rule: If DuplicateScore ≥ 0.72 , the report is linked to the primary incident.

7. Automated Priority & SLA Dynamic Matrix

Every complaint is automatically assigned an operational priority and binding Service Level Agreement (SLA) countdown based on machine learning risk scoring:

- **CRITICAL (4 Hours SLA):** Active gas leaks, fallen high-voltage power lines, massive water main breaches, bridge structural collapses.
- **HIGH (24 Hours SLA):** Major arterial road potholes, overflowing sewage on walkways, non-functioning traffic signals at major junctions.
- **MEDIUM (72 Hours SLA):** Residential street potholes, uncollected community bins, non-working streetlights in low-traffic alleys.
- **LOW (168 Hours / 7 Days SLA):** Cosmetic footpath cracks, park bench repainting, tree trimming, minor drainage silt accumulation.

8. Machine Learning Lifecycle & Admin Controls

The AI architecture is configurable via the Admin AI Configuration console (`/admin/ai-config`). Administrators can dynamically tune parameters in production:

- **Primary Engine Selector:** Toggle between Gemini 2.5 Flash, Gemini 1.5 Pro, and Offline Bayesian Statistical NLP.
- **Confidence Threshold Tuner:** Adjust auto-routing threshold (default 0.70) to balance precision vs automation.
- **Sampling Temperature:** Tune LLM temperature (0.0 for deterministic classification, 0.3 for creative reporting).
- **Continuous Retraining:** Re-ingests citizen and department verification outcomes to update Bayesian vocabulary weights.

9. Technical Architecture Summary Table

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